

Investment Financing Conservatism and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Investment Financing Conservatism (IFC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IFC achieves an annualized gross (net) Sharpe ratio of 0.59 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.18 (2.57), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth) is 20 bps/month with a t-statistic of 2.90.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remain central questions in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This gradual diffusion manifests particularly in the cross-section of stock returns, where firms with similar fundamental characteristics exhibit persistent return differences that cannot be explained by traditional risk factors.

Despite extensive research on information-based anomalies, the literature has paid limited attention to how firms' financing choices for their investments signal future profitability and how slowly this information incorporates into prices. Investment financing decisions represent fundamental corporate policy choices that directly affect firm value, yet investors may not immediately recognize their implications due to limited attention and processing constraints. This gap suggests that investment financing patterns may predict returns, particularly where information frictions are most severe.

We propose that Investment Financing Conservatism (IFC) predicts stock returns through the mechanism of slow information diffusion. When firms exhibit conservative financing patterns for their investments—relying less on external financing and maintaining financial flexibility—they signal stronger internal cash generation and better investment opportunities. However, investors with limited attention fail to immediately process these signals, leading to systematic underreaction in stock prices [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#).

The gradual diffusion of information framework predicts that value-relevant signals generate return continuation as information slowly incorporates into prices [Hong et al. \(2000a\)](#); [Hong and Stein \(1999\)](#). Conservative investment financing represents

precisely such a signal: it requires careful analysis of financial statements to identify and interpret, making it particularly susceptible to delayed recognition. Investors facing attention constraints prioritize salient information while overlooking subtle indicators embedded in financing patterns [Peng and Xiong \(2006\)](#). This selective attention creates temporary mispricing that corrects gradually as information diffuses through the market.

The underreaction should concentrate among stocks where information frictions are most severe. Small firms with low analyst coverage experience slower information diffusion because fewer sophisticated investors monitor and trade these securities [Hong et al. \(2000b\)](#). Additionally, retail investors, who dominate trading in neglected stocks, exhibit stronger behavioral biases and slower information processing [Barber and Odean \(2008\)](#). These cross-sectional predictions provide testable implications: IFC should predict stronger return continuation among small, low-coverage stocks where gradual information incorporation dominates price formation.

Our empirical analysis reveals that Investment Financing Conservatism strongly predicts cross-sectional stock returns. The value-weighted long-short IFC strategy generates an average monthly return of 29 basis points with a t-statistic of 4.26, translating to an annualized Sharpe ratio of 0.59. These returns remain economically and statistically significant after controlling for known risk factors: the strategy's alpha relative to the Fama-French six-factor model equals 22 basis points per month with a t-statistic of 3.18.

The predictive power of IFC exhibits remarkable robustness across alternative portfolio construction methodologies. Whether using name breaks, market capitalization breaks, or decile sorts, the strategy maintains statistical significance with t-statistics exceeding 3.00 in most specifications. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains 0.47, placing the strategy in the top 2% of all documented anomalies. The net alpha relative to

the six-factor model equals 17 basis points per month with a t-statistic of 2.57, confirming that IFC expands the investable opportunity set even after implementation costs.

Consistent with the gradual information diffusion hypothesis, the IFC effect concentrates where attention constraints bind most severely. Among the largest quintile of stocks—where information disseminates rapidly through analyst coverage and institutional trading—the strategy still generates 39 basis points per month with a t-statistic of 4.43. The alpha relative to the six-factor model for large-cap stocks equals 31 basis points with a t-statistic of 3.49. Controlling for six closely related anomalies including asset growth and net external financing, IFC maintains an alpha of 20 basis points per month with a t-statistic of 2.90, demonstrating that it captures distinct information about future returns.

This paper makes several contributions to the asset pricing and behavioral finance literatures. First, we document a novel predictor of cross-sectional returns that ranks in the top 5% of all anomalies in terms of Sharpe ratio, both gross and net of transaction costs. While [Cooper et al. \(2008\)](#) examine asset growth and [Bradshaw et al. \(2006\)](#) study external financing, our focus on the conservatism of investment financing provides a more nuanced signal that captures firms’ financial flexibility and internal capital generation. Unlike [Titman et al. \(2004\)](#) who examine aggregate investment, we specifically isolate the financing dimension of corporate investment decisions.

Methodologically, we employ the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to ensure our results withstand multiple robustness checks and avoid data-mining concerns. Our analysis spans over 50 years of data and survives controls for 212 documented anomalies, providing confidence that IFC represents a genuine pricing phenomenon rather than a statistical artifact. The strategy’s performance after transaction costs, verified using high-frequency effective spreads from [Chen](#)

and Velikov (2022), demonstrates its implementability for institutional investors. By showing that IFC maintains significance after controlling for related anomalies from the Journal of Financial Economics and Review of Financial Studies, we establish its incremental contribution to the factor zoo.

Our findings contribute to the broader literature on information diffusion and market efficiency. The concentration of returns among stocks with greater information frictions supports models of limited investor attention and gradual information incorporation Merton (1987). The persistence of the IFC effect, even among large-cap stocks with substantial analyst coverage, suggests that sophisticated investors also face constraints in processing complex accounting signals. These results have implications for market efficiency, corporate finance, and investment management, demonstrating that fundamental analysis of financing patterns continues to provide alpha even in increasingly efficient markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 6000-6999 and 4900-4999) following standard practice in the corporate finance literature. Investment Financing Conservatism signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities with maturities exceeding one year. This item reflects the gross amount of long-term debt raised before considering any repayments or retirements. Capital expenditure (CAPX) measures the funds used for

additions to property, plant, and equipment, excluding acquisitions. This variable captures the firm’s investment in tangible productive assets and represents a critical component of corporate investment policy. We construct the Investment Financing Conservatism signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital expenditure: $(\text{DLTIS}(t) - \text{DLTIS}(t-1)) / \text{CAPX}(t-1)$. This ratio captures the change in external debt financing relative to the firm’s recent investment intensity. A higher value indicates more aggressive debt financing growth relative to capital investment levels, while a lower or negative value suggests conservative financing behavior where debt issuance growth lags behind or declines relative to capital spending. We calculate this measure annually using fiscal year-end values and require non-missing observations for both DLTIS in consecutive years and CAPX in the prior year. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative capital expenditure in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFC signal. Panel A plots the time-series of the mean, median, and interquartile range for IFC. On average, the cross-sectional mean (median) IFC is -3.65 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IFC data. The signal’s interquartile range spans -3.29 to 4.32. Panel B of Figure 1 plots the time-series of the coverage of the IFC signal for the CRSP universe. On average, the IFC signal is available for 6.07% of CRSP names, which on average make up 7.19% of total market capitalization.

4 Does IFC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFC portfolio and sells the low IFC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short IFC strategy earns an average return of 0.29% per month with a t-statistic of 4.26. The annualized Sharpe ratio of the strategy is 0.59. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 6.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 504 stocks and an average market capitalization of at least \$1,502 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.18. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-25bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the IFC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFC, as well as average returns and al-

phas for long/short trading IFC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IFC strategy achieves an average return of 39 bps/month with a t-statistic of 4.43. Among these large cap stocks, the alphas for the IFC strategy relative to the five most common factor models range from 31 to 45 bps/month with t-statistics between 3.49 and 5.02.

5 How does IFC perform relative to the zoo?

Figure 2 puts the performance of IFC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IFC strategy falls in the distribution. The IFC strategy’s gross (net) Sharpe ratio of 0.59 (0.47) is greater than 95% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFC strategy (red line).² Ignoring trading costs, a \$1 invested in the IFC strategy would have yielded \$4.37 which ranks the IFC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFC strategy would have yielded \$2.73 which ranks the IFC strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the IFC relative to those. Panel A shows that the IFC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFC strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFC ranks between the 79 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IFC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of IFC with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFC or at least to weaken the power IFC has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC}IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFC signal in these Fama-MacBeth regressions exceed 1.76, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on IFC is 0.71.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFC strategy earns alphas that range from 19-22bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.86, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFC trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.90.

7 Does IFC add relative to the whole zoo?

Finally, we can ask how much adding IFC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IFC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes IFC grows to \$1241.53.

8 Conclusion

This study provides compelling evidence that Investment Financing Conservatism (IFC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IFC is available.

Novy-Marx and Velikov (2023), demonstrates that IFC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IFC delivers an annualized Sharpe ratio of 0.59 gross and 0.47 net of transaction costs, indicating strong economic viability even in practical implementation. The strategy’s resilience is particularly noteworthy when examined against the Fama-French five-factor model augmented with momentum, where it generates statistically significant alphas of 22 basis points per month (t -statistic = 3.18) on a gross basis and 17 basis points net of costs (t -statistic = 2.57). Perhaps most importantly, the signal maintains its predictive power even after controlling for six closely related financing and investment strategies, yielding a gross alpha of 20 basis points per month with a t -statistic of 2.90. This orthogonality to related factors suggests that IFC captures unique information about firm fundamentals not fully reflected in existing asset pricing models.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, IFC emerges as a distinct source of cross-sectional return variation that warrants inclusion in factor models and asset pricing tests. For practitioners, the signal’s robust performance after transaction costs and its independence from related strategies suggest potential value as either a standalone strategy or as a complement to existing factor-based investment approaches. The persistence of returns after controlling for multiple factors indicates that markets may systematically misprice firms based on their financing conservatism, potentially due to behavioral biases or limits to arbitrage.

Several limitations of this study merit consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across

all market regimes. Second, our transaction cost estimates, while conservative, may not fully account for market impact costs associated with large-scale implementation. Third, the analysis focuses primarily on U.S. equity markets, and the generalizability of IFC to international markets remains unexplored.

Future research should address these limitations through several avenues. International studies could examine whether IFC's predictive power extends to developed and emerging markets with different institutional frameworks and investor bases. Time-varying analysis could investigate whether the IFC premium fluctuates with market conditions, sentiment, or macroeconomic variables. Additionally, research into the economic mechanisms underlying the IFC effect would be valuable—specifically, whether the premium reflects compensation for systematic risk, behavioral mispricing, or institutional frictions. Finally, examining the interaction between IFC and corporate governance, information asymmetry, or financial constraints could provide deeper insights into why conservative financing policies predict future returns. Understanding these mechanisms would not only enhance our theoretical understanding of asset pricing but also inform more effective investment strategies and corporate financial policies.

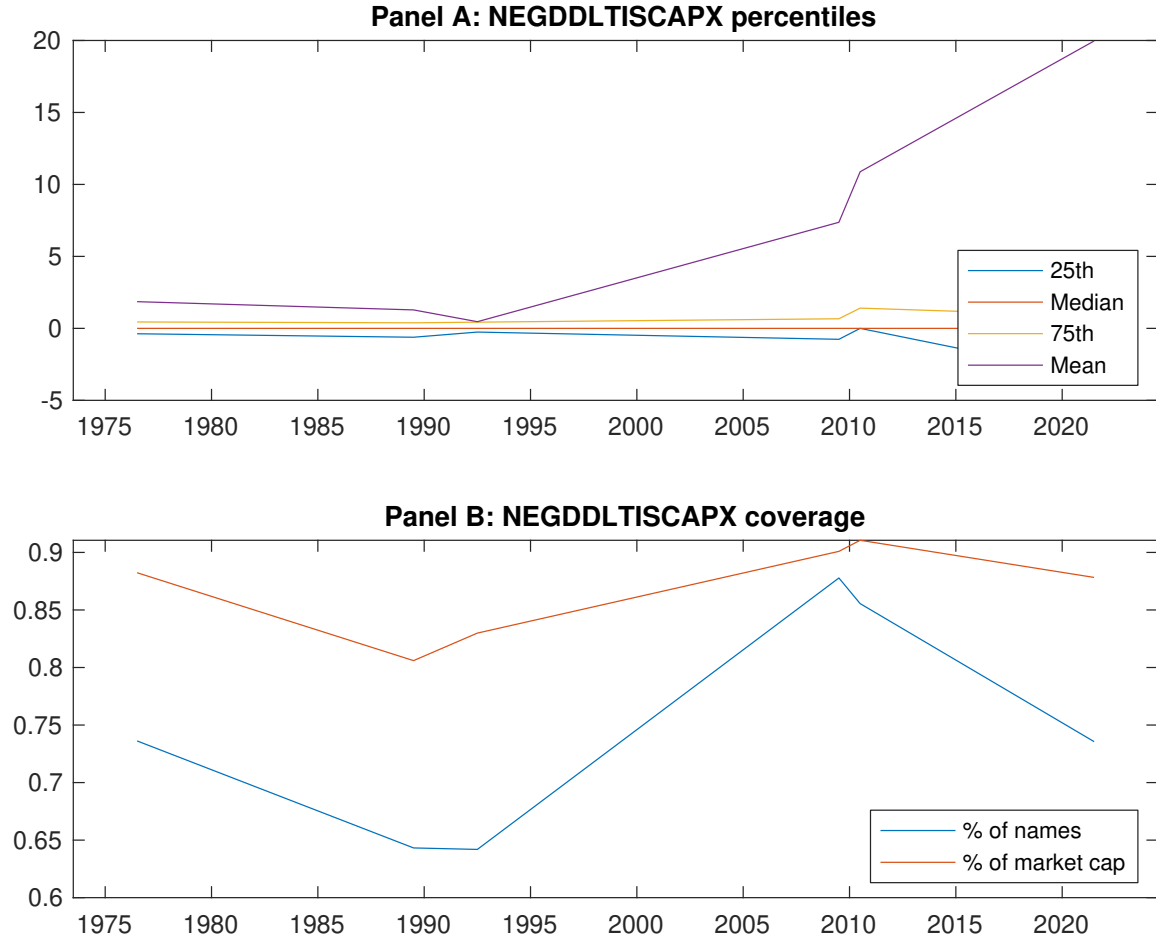


Figure 1: Times series of IFC percentiles and coverage. This figure plots descriptive statistics for IFC. Panel A shows cross-sectional percentiles of IFC over the sample. Panel B plots the monthly coverage of IFC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IFC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.56]	0.67 [3.76]	0.65 [3.26]	0.72 [4.05]	0.84 [4.19]	0.29 [4.26]
α_{CAPM}	-0.18 [-3.44]	0.07 [1.35]	-0.02 [-0.33]	0.12 [2.35]	0.16 [3.00]	0.34 [5.05]
α_{FF3}	-0.20 [-3.91]	0.04 [0.97]	0.05 [0.93]	0.12 [2.38]	0.13 [2.57]	0.34 [4.95]
α_{FF4}	-0.17 [-3.21]	0.04 [0.91]	0.10 [1.86]	0.07 [1.44]	0.13 [2.40]	0.30 [4.30]
α_{FF5}	-0.16 [-3.10]	-0.03 [-0.77]	0.10 [1.75]	0.03 [0.65]	0.08 [1.48]	0.24 [3.54]
α_{FF6}	-0.14 [-2.65]	-0.03 [-0.68]	0.13 [2.41]	0.00 [0.06]	0.08 [1.47]	0.22 [3.18]
Panel B: Fama and French (2018) 6-factor model loadings for IFC-sorted portfolios						
β_{MKT}	1.08 [89.35]	0.98 [95.69]	0.97 [75.06]	0.96 [83.22]	1.05 [84.68]	-0.03 [-1.99]
β_{SMB}	0.11 [6.08]	-0.12 [-7.63]	-0.02 [-0.83]	-0.02 [-1.12]	0.11 [6.04]	0.00 [0.09]
β_{HML}	0.08 [3.42]	0.06 [2.92]	-0.17 [-6.82]	-0.08 [-3.57]	-0.02 [-0.83]	-0.10 [-3.28]
β_{RMW}	0.01 [0.36]	0.14 [7.09]	-0.04 [-1.40]	0.08 [3.53]	0.08 [3.09]	0.07 [2.15]
β_{CMA}	-0.16 [-4.48]	0.10 [3.20]	-0.10 [-2.76]	0.22 [6.55]	0.13 [3.62]	0.29 [6.28]
β_{UMD}	-0.04 [-3.03]	-0.01 [-0.59]	-0.06 [-4.54]	0.05 [4.10]	-0.00 [-0.02]	0.04 [2.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	504	1021	562	636	
me (\$10 ⁶)	1502	2808	2233	2766	1568	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IFC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29 [4.26]	0.34 [5.05]	0.34 [4.95]	0.30 [4.30]	0.24 [3.54]	0.22 [3.18]
Quintile	NYSE	EW	0.19 [4.18]	0.21 [4.65]	0.20 [4.38]	0.19 [4.13]	0.18 [4.01]	0.18 [3.91]
Quintile	Name	VW	0.26 [3.58]	0.31 [4.26]	0.33 [4.46]	0.29 [3.88]	0.26 [3.63]	0.24 [3.31]
Quintile	Cap	VW	0.31 [4.74]	0.35 [5.46]	0.36 [5.68]	0.31 [4.82]	0.28 [4.41]	0.25 [3.87]
Decile	NYSE	VW	0.32 [3.64]	0.37 [4.16]	0.37 [4.21]	0.36 [4.04]	0.24 [2.78]	0.25 [2.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [3.36]	0.28 [4.07]	0.27 [3.93]	0.25 [3.61]	0.19 [2.81]	0.17 [2.57]
Quintile	NYSE	EW	-0.08 [-1.37]					
Quintile	Name	VW	0.20 [2.74]	0.24 [3.28]	0.25 [3.44]	0.23 [3.14]	0.21 [2.80]	0.19 [2.57]
Quintile	Cap	VW	0.25 [3.92]	0.29 [4.48]	0.30 [4.65]	0.27 [4.22]	0.23 [3.59]	0.20 [3.28]
Decile	NYSE	VW	0.25 [2.79]	0.29 [3.21]	0.29 [3.23]	0.29 [3.17]	0.18 [2.06]	0.18 [2.02]

Table 3: Conditional sort on size and IFC

This table presents results for conditional double sorts on size and IFC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFC and short stocks with low IFC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IFC Quintiles					IFC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.72 [2.65]	0.90 [3.26]	0.91 [3.25]	0.95 [3.32]	0.77 [2.84]	0.05 [0.70]	0.07 [0.90]	0.05 [0.70]	0.05 [0.65]	0.08 [1.00]	0.08 [0.97]
	(2)	0.81 [3.06]	0.81 [3.20]	0.87 [3.41]	0.91 [3.64]	0.81 [3.24]	0.01 [0.08]	0.04 [0.52]	0.00 [0.04]	0.01 [0.16]	-0.04 [-0.50]	-0.03 [-0.36]
	(3)	0.74 [3.01]	0.85 [3.76]	0.80 [3.30]	0.79 [3.48]	0.93 [3.98]	0.19 [2.57]	0.23 [3.17]	0.22 [3.03]	0.18 [2.37]	0.21 [2.85]	0.18 [2.38]
	(4)	0.74 [3.34]	0.80 [3.76]	0.82 [3.71]	0.78 [3.63]	0.88 [4.07]	0.14 [1.88]	0.16 [2.19]	0.14 [1.93]	0.12 [1.68]	0.09 [1.26]	0.09 [1.15]
	(5)	0.45 [2.20]	0.63 [3.55]	0.59 [3.05]	0.62 [3.41]	0.84 [4.34]	0.39 [4.43]	0.42 [4.78]	0.45 [5.02]	0.37 [4.21]	0.36 [4.01]	0.31 [3.49]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IFC Quintiles					IFC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	380	381	381	381	377	35	34	31	33	34	
	(2)	105	105	105	105	105	60	60	59	60	59	
	(3)	75	75	75	75	75	106	106	103	104	106	
	(4)	63	63	63	63	62	226	234	225	230	224	
(5)	57	57	57	57	57	1328	2082	1748	2145	1446		

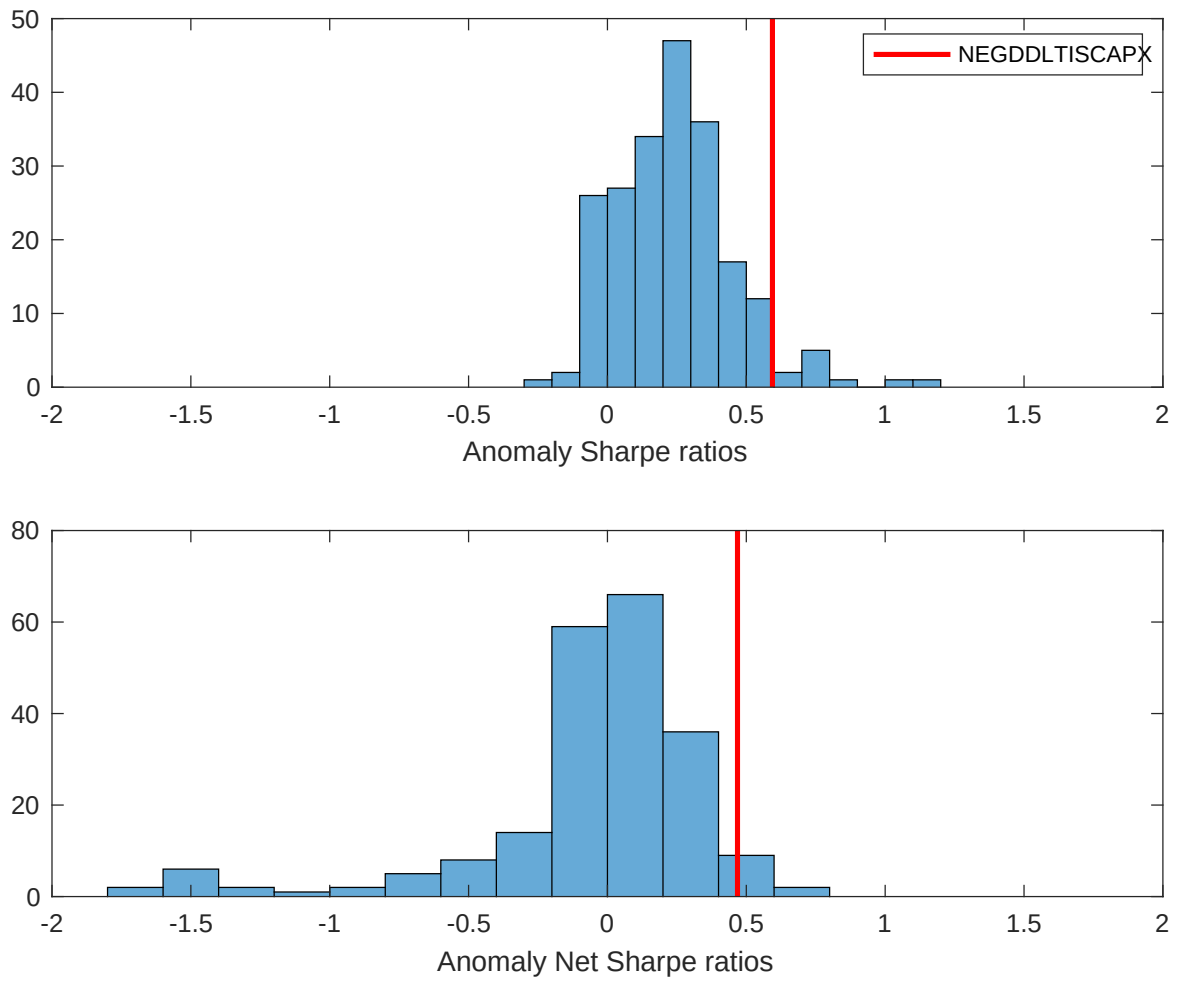


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

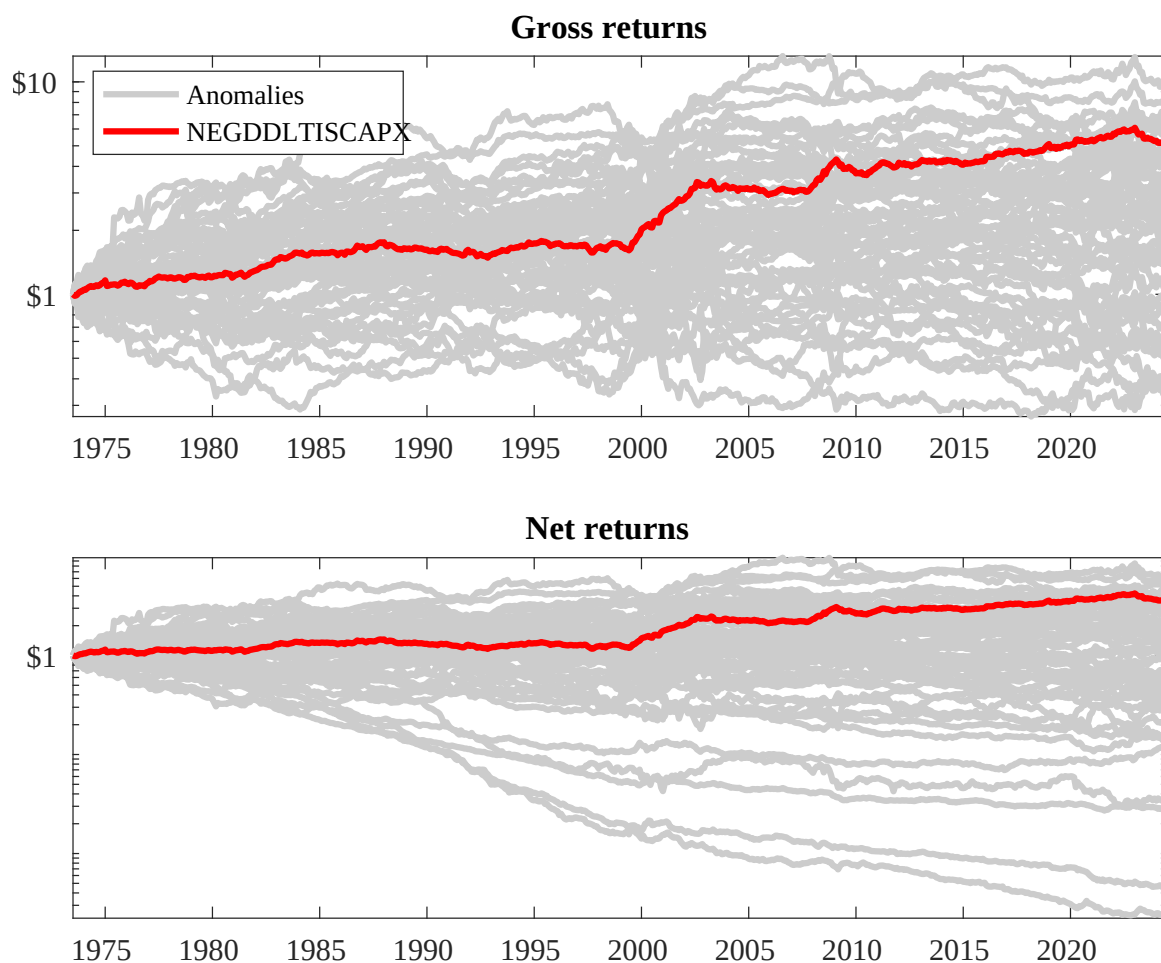
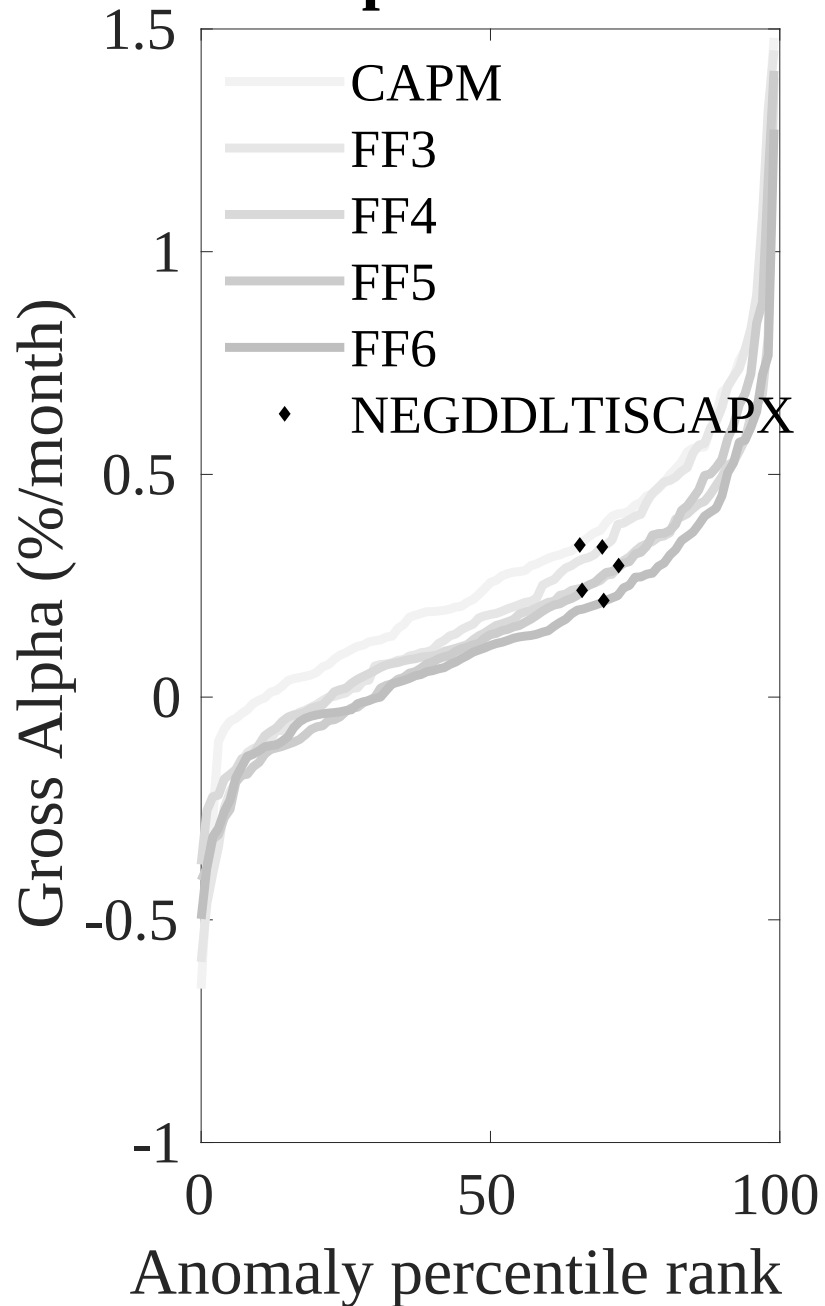


Figure 3: Dollar invested.

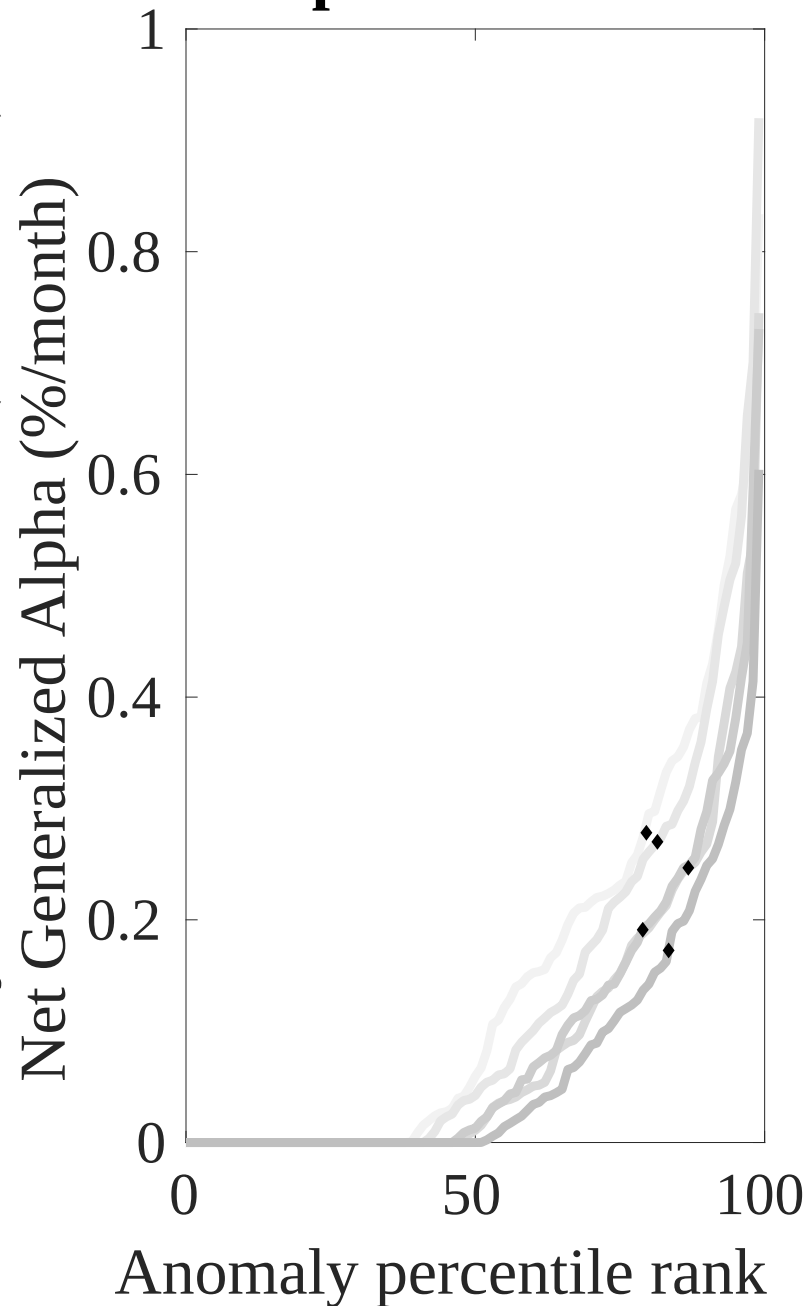
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



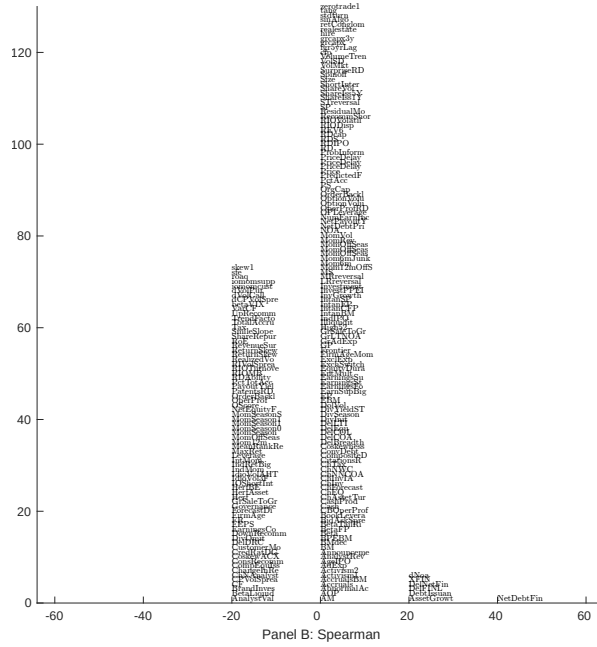
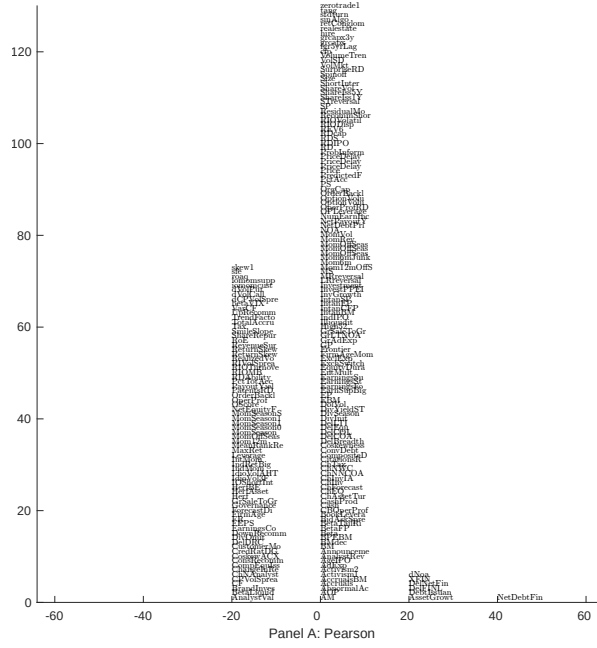


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with IFC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

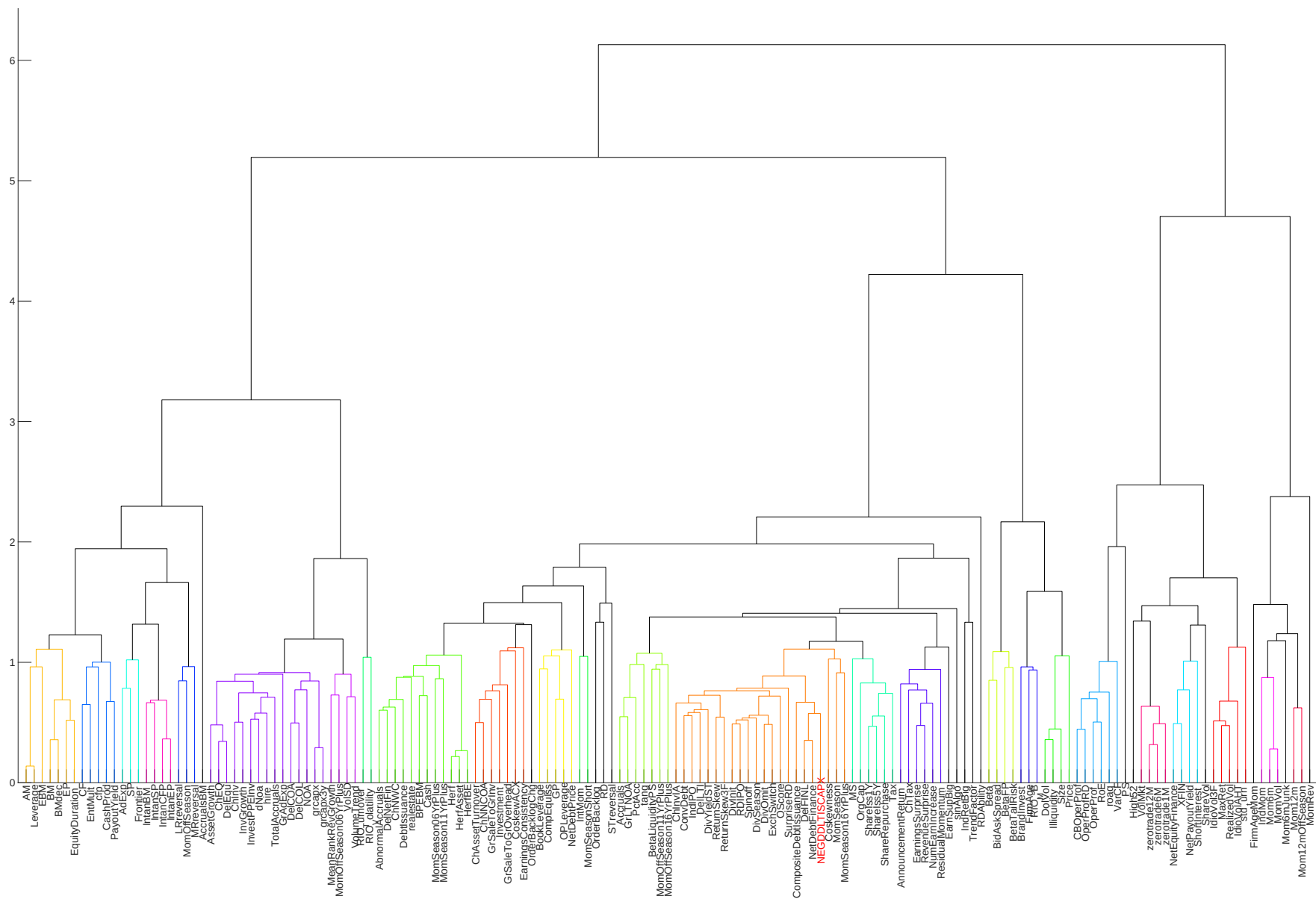


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

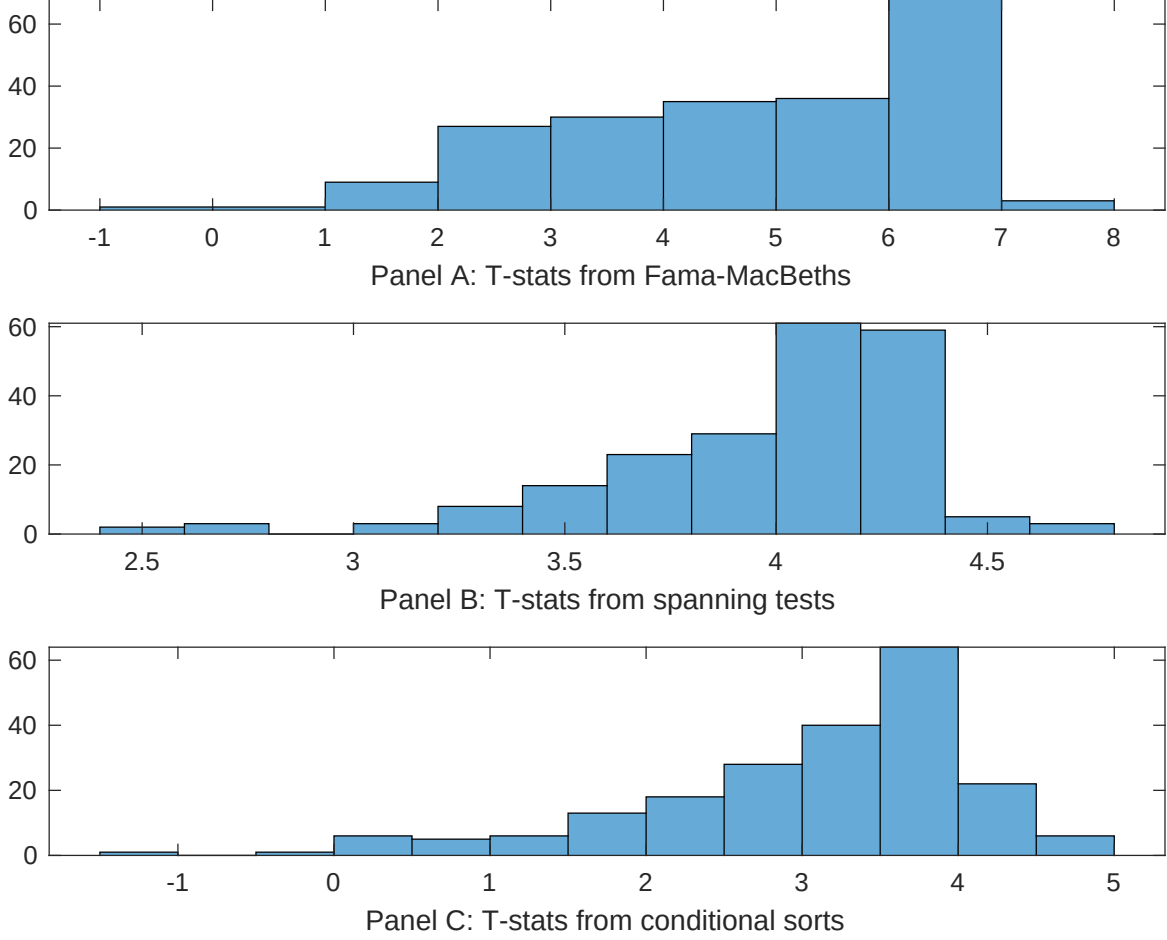


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC}IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IFC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 1973Q6 to 2024Q6.

Intercept	0.13 [5.38]	0.14 [5.69]	0.13 [5.34]	0.14 [5.81]	0.14 [5.68]	0.13 [5.35]	0.15 [5.96]
IFC	0.51 [2.10]	0.80 [3.11]	0.61 [2.32]	0.54 [2.05]	0.45 [1.76]	0.13 [5.28]	0.19 [0.71]
Anomaly 1	0.17 [9.10]						-0.54 [-1.32]
Anomaly 2		0.20 [6.43]					0.80 [1.53]
Anomaly 3			0.21 [9.11]				0.60 [1.06]
Anomaly 4				0.11 [8.61]			0.49 [2.64]
Anomaly 5					0.14 [9.67]		0.73 [4.55]
Anomaly 6						0.95 [6.20]	0.12 [0.98]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IFC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19 [2.86]	0.20 [2.89]	0.20 [2.93]	0.21 [3.05]	0.21 [3.01]	0.22 [3.18]	0.20 [2.90]
Anomaly 1	14.96 [3.91]						11.09 [2.10]
Anomaly 2		10.79 [3.18]					7.41 [2.02]
Anomaly 3			15.30 [4.15]				7.07 [1.40]
Anomaly 4				8.20 [2.08]			6.10 [1.41]
Anomaly 5					1.70 [0.43]		-7.28 [-1.66]
Anomaly 6						3.45 [0.94]	2.09 [0.54]
mkt	-2.57 [-1.65]	-1.43 [-0.88]	-2.88 [-1.86]	-2.91 [-1.86]	-2.91 [-1.85]	-2.82 [-1.79]	-1.59 [-0.97]
smb	-0.73 [-0.31]	4.27 [1.64]	-0.55 [-0.23]	0.21 [0.09]	0.84 [0.35]	1.04 [0.43]	0.99 [0.36]
hml	-8.39 [-2.79]	-8.18 [-2.70]	-8.76 [-2.93]	-9.77 [-3.23]	-9.51 [-3.12]	-10.04 [-3.22]	-7.62 [-2.45]
rmw	5.31 [1.73]	0.21 [0.06]	4.97 [1.62]	6.68 [2.18]	6.68 [2.17]	6.81 [2.21]	0.45 [0.12]
cma	23.78 [5.04]	21.75 [4.29]	24.82 [5.35]	18.86 [2.85]	27.52 [5.08]	25.70 [4.55]	14.55 [2.08]
umd	2.47 [1.56]	3.69 [2.38]	2.78 [1.78]	3.93 [2.52]	3.71 [2.37]	3.54 [2.24]	2.52 [1.59]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	13	13	14	12	11	11	15

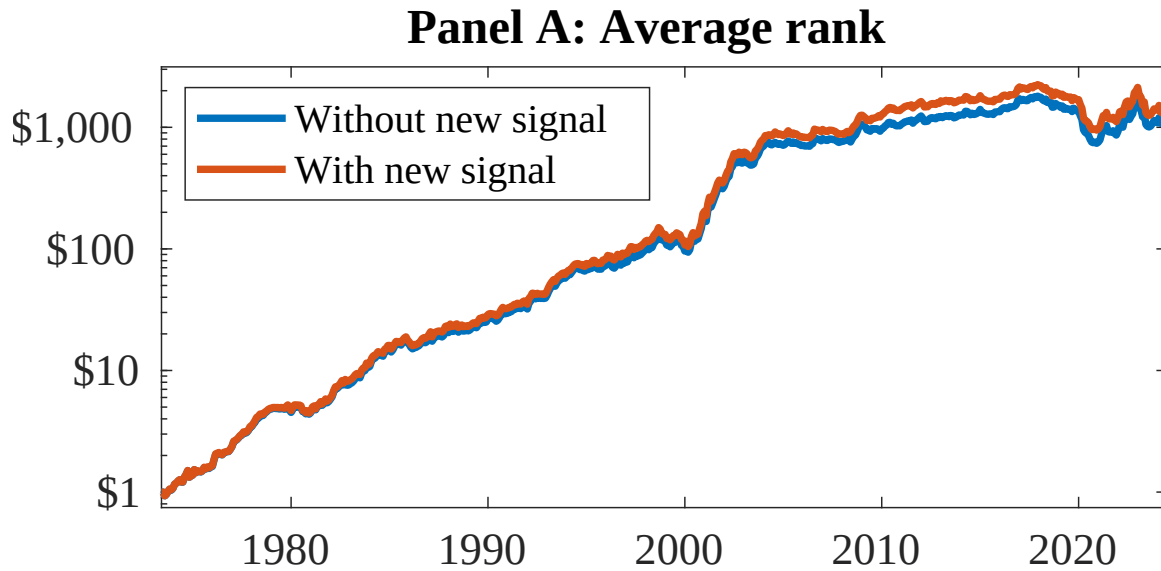


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IFC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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